THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE UNIT TEST SET A

Chapters 1–4: **Exploring the Investigative World of Science, The Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. In Grade 6 the book began with: 1

(a) wonder through why and how questions **(b)** only experiments
(c) only formulas **(d)** only history

2. Which of these can be changed to study puri puffing? 1

(a) temperature of the oil **(b)** size of the kitchen
(c) colour of the plate **(d)** name of the cook

3. Keeping notes of everything seen, smelt and sensed during an experiment helps in: 1

(a) changing the results **(b)** skipping measurements
(c) making sense of the results later **(d)** avoiding the experiment

4. A bacterial cell lacks a well-defined: 1

(a) cell wall **(b)** cell membrane **(c)** nucleus **(d)** cytoplasm

5. Biogas is mainly produced by microbes acting on: 1

(a) glass **(b)** iron
(c) sand **(d)** cow dung and organic waste

6. Rhizobium lives in the root nodules of: 1

(a) cereals **(b)** legumes **(c)** grasses **(d)** cactus

7. A vaccine works by: 1

(a) curing the disease at once **(b)** supplying vitamins
(c) training the immune system **(d)** killing all bacteria

8. Anaemia is mainly due to lack of: 1

(a) iron **(b)** iodine **(c)** vitamin C **(d)** calcium

9. Which of these is a communicable disease? 1

(a) Diabetes **(b)** Scurvy **(c)** Typhoid **(d)** Goitre

10. Heating effect of current is due to the: 1

(a) length of the cell **(b)** magnetism of the wire
(c) resistance of the conductor **(d)** colour of the wire

11. The electrolyte in a dry cell is a: 1

(a) dry powder only **(b)** moist paste **(c)** metal rod **(d)** gas

12. A voltaic cell has two different metal rods and a: 1

(a) magnet **(b)** insulator **(c)** electrolyte **(d)** bulb

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. State why only one thing should be changed at a time. 2

14. What is a nucleoid? In which cells is it found? 2

15. How does Rhizobium help farmers? 2

16. Define communicable and non-communicable diseases. 2

17. What is an electromagnet? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Explain why the puffing of a puri is said to be 'not completely understood'. 3

19. Describe how biogas is made and used. 3

20. Differentiate communicable and non-communicable diseases in a table form with two points and two examples. 3

21. State the factors that affect the strength of an electromagnet with the effect of each. 3

22. List the factors on which heat produced in a wire depends. 3

OR

Describe three applications of the heating effect of electric current. 3

23. Why do scientists keep asking new questions after an answer? 3

OR

How does the chapter link the Moon's phases to calendars? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

24. Why are cell shapes related to their functions? Describe three types of cells to explain. 5

OR

Explain the biogas process and its two advantages. 5

25. Describe the structure and working of a dry cell with a labelled diagram. 5

OR

Explain how the lemon cell activity is done and what it teaches about voltaic cells. 5

END OF QUESTION PAPER Total: 50 Marks

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